

Information Systems - MERISE Modeling Report

Department of Computer Engineering | 2nd Year Level

Exercise N° 5: School Administration System

1. Attribute Justification (STUDENT Entity)

Attribute	Scientific Justification
Student_ID	Primary Key (PK). Ensures unique tuple identification.
First_Name, Last_Name	Personal identifiers for administrative consistency.
Date_of_Birth	Calculated attribute for age constraints and eligibility.
Address, Phone, Email	Communication link between the administration and the subject.
Class_Code	Foreign Key (FK). Materializes the 1:N relationship with CLASS.

2. Conceptual Data Model (MCD)

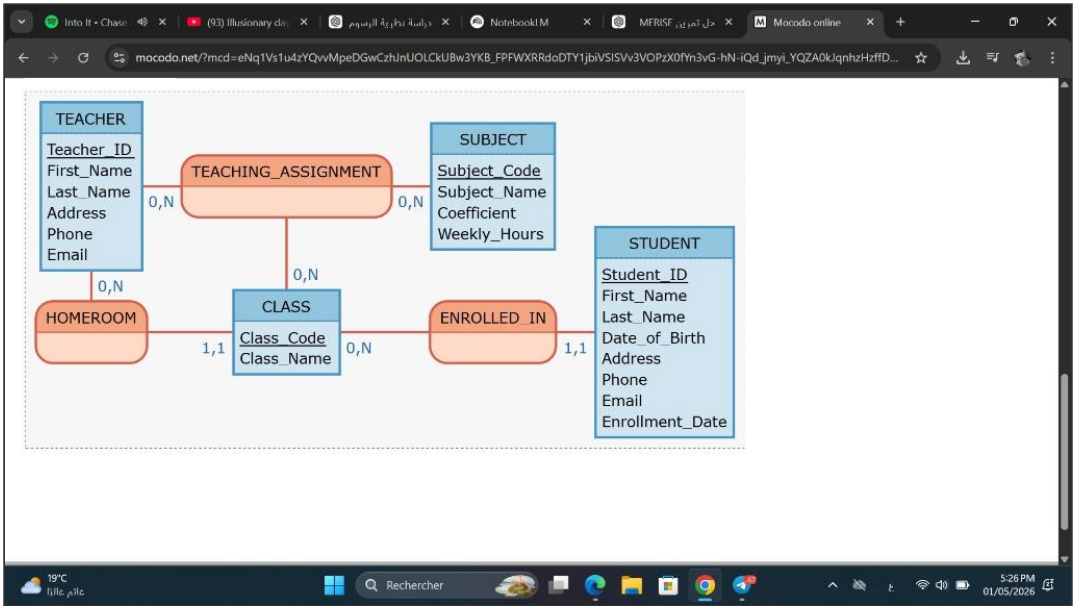


Figure 1: Conceptual Data Model for Exercise 5

3. Logical Data Model (LDM / MLD)

STUDENT(<u>Student_ID</u> , First_Name, Last_Name, Date_of_Birth, Address, Phone, Email, Enrollment_Date, #Class_Code)
CLASS(<u>Class_Code</u> , Class_Name, #Homeroom_Teacher_ID)

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TEACHER(Teacher_ID, First_Name, Last_Name, Address, Phone, Email)
SUBJECT(Subject_Code, Subject_Name, Coefficient, Weekly_Hours)
TEACHING_ASSIGNMENT(#Teacher_ID, #Class_Code, #Subject_Code)

```

Exercise N° 6: Student Evaluation Management

1. Conceptual Data Model (MCD)

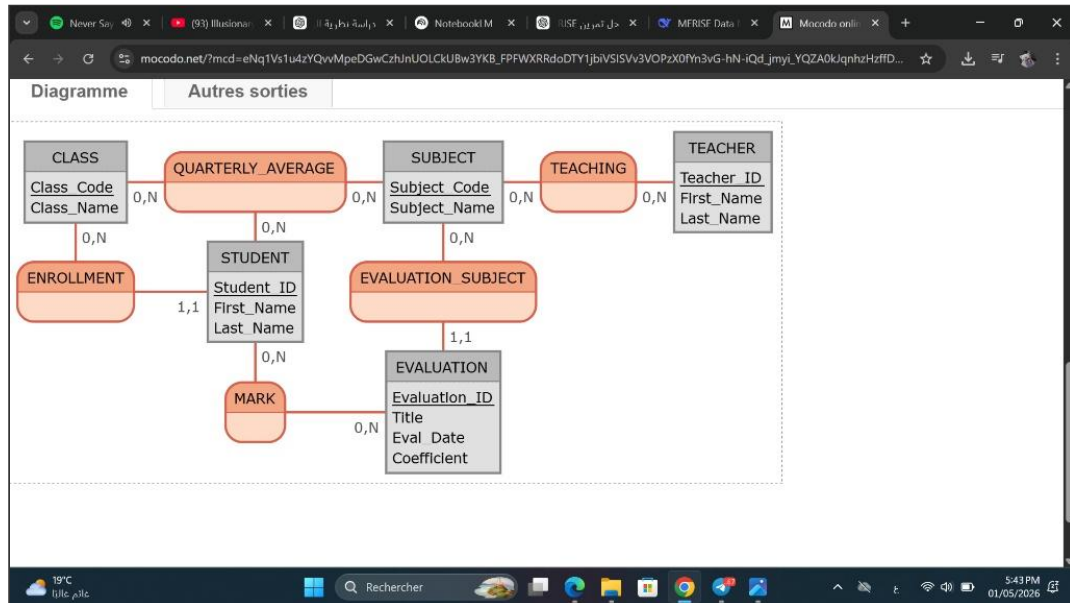


Figure 2: Conceptual Data Model for Evaluations

2. Logical Data Model (LDM / MLD)

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STUDENT(Student_ID, First_Name, Last_Name, #Class_Code)
CLASS(Class_Code, Class_Name)
TEACHER(Teacher_ID, First_Name, Last_Name)
SUBJECT(Subject_Code, Subject_Name)
EVALUATION(Evaluation_ID, Title, Date, Coefficient, #Subject_Code)
MARK(#Student_ID, #Evaluation_ID, Mark_Value)
QUARTERLY_AVERAGE(#Student_ID, #Subject_Code, #Class_Code, Quarter,
Average)

```

Exercise N° 7: Extended Faculty Assignment

1. Conceptual Data Model (MCD) Extension

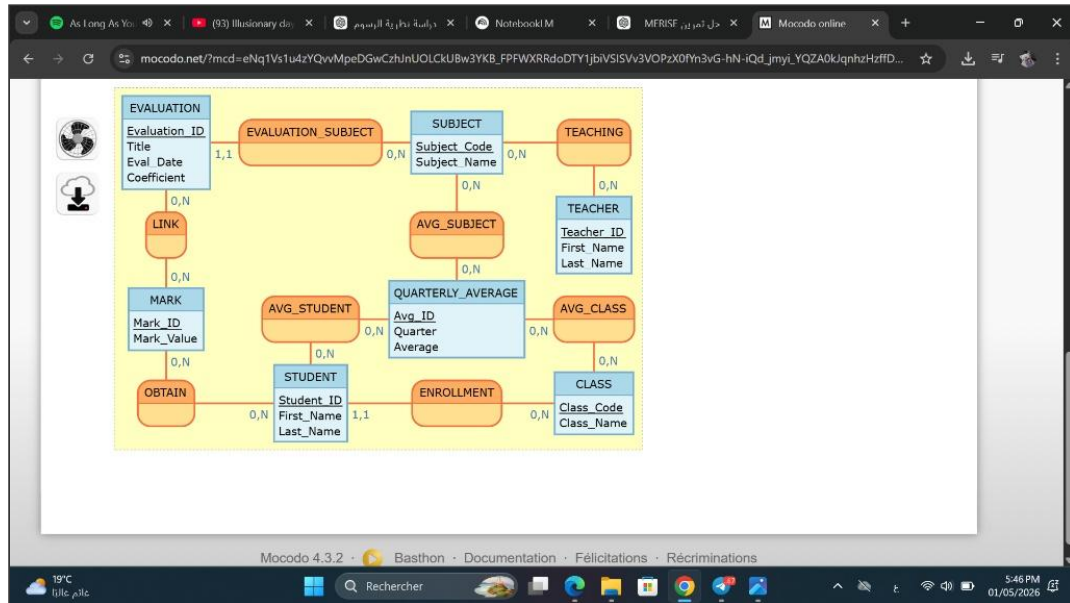


Figure 3: Extended Model for Class-Teacher Assignment

2. Logical Data Model (LDM / MLD)

```
STUDENT(Student_ID, First_Name, Last_Name, #Class_Code)
CLASS(Class_Code, Class_Name)
TEACHER(Teacher_ID, First_Name, Last_Name)
SUBJECT(Subject_Code, Subject_Name)
CLASS_TEACHER(#Teacher_ID, #Class_Code)
TEACHER_SUBJECT(#Teacher_ID, #Subject_Code)
EVALUATION(Evaluation_ID, Title, Eval_Date, Coefficient,
#Subject_Code)
MARK(#Student_ID, #Evaluation_ID, Mark_Value)
QUARTERLY_AVERAGE(#Student_ID, #Subject_Code, #Class_Code, Quarter,
Average)
```

[End of Official Documentation]